Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2019-P2-Model)** **Date: 22-03-2026**

Time: 02.30Pm to 05.30Pm **GTA-19** **Max. Marks: 186**

KEY SHEET

PHYSICS

1	**ABCD**	2	**BD**	3	**ABC**	4	**ACD**	5	**BC**	6	**D**
7	**AD**	8	**AB**	9	**4**	10	**73 – 75**	11	**0.5**	12	**1**
13	**12**	14	**0.25**	15	**D**	16	**A**	17	**D**	18	**B**

CHEMISTRY

19	**CD**	20	**CD**	21	**ABC**	22	**BC**	23	**ABD**	24	**ABC**
25	**BC**	26	**ABC**	27	**14**	28	**2**	29	**130**	30	**333.33**
31	**24**	32	**1299**	33	**D**	34	**C**	35	**C**	36	**A**

MATHEMATICS

37	**BCD**	38	**BC**	39	**ACD**	40	**ABC**	41	**AC**	42	**BD**
43	**ABCD**	44	**ABD**	45	**4**	46	**164**	47	**10.66 – 10.67**	48	**2**
49	**9.5**	50	**6**	51	**B**	52	**D**	53	**A**	54	**D**

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SOLUTIONS

PHYSICS

1. For free (uncoupled) circular orbits: $v_1' = \sqrt{\frac{GM}{R_1}},\ v_2' = \sqrt{\frac{GM}{R_2}}.$

When the satellites are connected by a tensioned cable, the centripetal forces become:

$$\frac{mv_1^2}{R_1} = \frac{GMm}{R_1^2} - T_c,$$

$$\frac{mv_2^2}{R_2} = \frac{GMm}{R_2^2} + T_c,$$

where T_c is the cable tension.

Both satellites must complete one revolution in the same orbital period :

$$v_1 = \frac{2\pi R_1}{T},\ v_2 = \frac{2\pi R_2}{T}.$$

$$\frac{4\pi^2}{T^2}(R_1 + R_2) = GM\left(\frac{R_2^2 + R_1^2}{R_1^2 R_2^2}\right).$$

Substitute $R_2 = 2R_1$:

2. If product has higher B/A then energy will be released.

A) $1 < A < 50 \longrightarrow$ Resultant has $A < 100$ i.e. B/A is same

B) $51 < A < 100 \longrightarrow$ Resultant has $100 < A < 200$ i.e. B/A increases so energy is released.

C) $100 < A < 200 \longrightarrow$ Fission results in each nuclei with $50 < A < 100$ so B/A decreases so no energy released.

D) $200 < A < 260 \longrightarrow$ Fission results in each nuclei with $100 < A < 130$ i.e. B/A increases so energy is released.

3. In equilibrium

$4mg\sin\theta = kx_0$

$k = 120{,}000$ N / m

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{2\ m}} = \frac{\sqrt{60}}{2\pi}\ Hz$$

New amplitude $= 20/2 = 10$ cm

4. Amplitude $= \frac{13}{2}$ m, $\omega = \frac{2\pi}{T} = \frac{\pi}{4}$ s^{-1}

$$\vec{r} = \frac{13}{2}(1-\cos\omega t)\frac{12}{13}\hat{i} + \left[5 - \frac{13}{2}(1-\cos\omega t)\frac{5}{13}\right]\hat{j}$$

$$\vec{r} = 12\sin^2\left(\frac{\pi t}{8}\right)\hat{i} + 5\cos^2\left(\frac{\pi t}{8}\right)\hat{j}$$

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$$\therefore \vec{v} = \left[\frac{3\pi}{2}\sin\left(\frac{\pi t}{4}\right)\hat{i} - \frac{5\pi}{8}\sin\left(\frac{\pi t}{4}\right)\hat{j}\right] \text{m/s}$$

5. Magnetic flux through the smaller loop due to current in the bigger loop

$$= \left(\frac{\mu_0 I}{2\pi\, d} - \frac{\mu_0 I}{2\pi(2\,d)}\right)\ell^2 \quad = \frac{\mu_0 I \ell^2}{4\pi\, d}$$

So mutual inductance $= \dfrac{\mu_0 \ell^2}{4\pi\, d}$

$$\therefore \text{So emf induced in the bigger loop} = \frac{d}{dt}\left[\frac{\mu_0 \ell^2}{4\pi\, d} kt\right] \qquad = \frac{\mu_0 \ell^2 k}{4\pi d}$$

6. Total charge (finally) = 0

So charge flown to the earth $= \sum_{i=1}^{N} Q_i$

7. $\dfrac{h}{4} = \dfrac{3h}{4}\sin\theta \;\; 1 = 3\sin\theta$

$$\sin\theta = \frac{1}{3} \qquad \theta = \sin^{-1}\left(\frac{1}{3}\right)$$

8. Before opening the valve:

- Gas state: p, V, T, amount n.
- Internal energy: $E_1 = nc_V T$.

Work Student did in compressing the gas: $A_1 = pV$

After the balloon is completely filled:

- Gas state: p_0, V_2, T_2 (same n).
- Internal energy: $E_2 = nc_V T_2$.

During inflation, the gas does work against outside pressure:

$$A_2 = p_0 V_2$$

Energy conservation for the gas:

$$E_2 - E_1 = A_1 - A_2$$

$$nc_V T_2 - nc_V T = pV - p_0 V_2$$

Use ideal gas equation for initial and final states:

$$pV = nRT,\; p_0 V_2 = nRT_2$$

Substitute:

$$nc_V (T_2 - T) = nRT - nRT_2 = nR(T - T_2)$$

$$\Rightarrow (c_V + R)(T_2 - T) = 0$$

Since $c_V + R = c_P \neq 0$,

$$T_2 = T$$

9. Opposite sides → forces cancel → uniform motion: $t_1 = \dfrac{2L}{v}$.

Both charges in same region → net force $2qE_0$:

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$$a=\frac{2qE_0}{m},\ 2v=at\Rightarrow t=\frac{mv}{E_0q}.$$

Occurs twice ⇒ $t_2=\frac{2mv}{E_0q}.$

Period $T=\frac{2L}{v}+\frac{2mv}{E_0q}$

10. Mass of water:

$$m=\rho V=2\text{ kg}.$$

Heat required to warm water from -3°C to 0°C:

$$Q=mc\Delta T=2\times4200\times3=25200\text{ J}.$$

If m_{ice} freezes:

$$m_{ice}\lambda=Q$$

$$m_{ice}=\frac{25200}{340000}\approx0.074\text{ kg}=74.11\text{ gm}$$

11. Capillary rise in the inner tube (radius r_1):

$$h=\frac{2\sigma\cos\theta}{\rho g r_1}.$$

For the annular space between radii r_1 and r_2 :

- Upward force by surface tension

$$F_{\uparrow}=2\pi\sigma\cos\theta(r_1+r_2)$$

- Weight of water column of height h:

$$F_{\downarrow}=\rho gh\pi\left(r_2^2-r_1^2\right).$$

Equating:

$$2\pi\sigma\cos\theta(r_1+r_2)=\rho gh\pi\left(r_2^2-r_1^2\right).$$

Substitute the same h as above (levels equal):

$$\frac{2\sigma\cos\theta}{\rho g r_1}=\frac{2\sigma\cos\theta(r_1+r_2)}{\rho g\left(r_2^2-r_1^2\right)}.$$

Cancel common factors:

$$\frac{1}{r_1}=\frac{r_1+r_2}{r_2^2-r_1^2}\Rightarrow r_2^2-r_1r_2-2r_1^2=0\Rightarrow r_2=2r_1$$

$$r_2=2r_1$$

12. For a single ring to get $\int\vec{B}.d\vec{x}$ complete the loop.

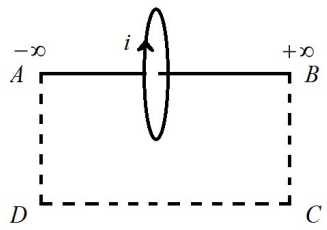
$$\underset{A}{\overset{B}{\int}}\vec{B}\cdot d\vec{x}+\underset{B}{\overset{C}{\int}}\vec{B}\cdot d\vec{x}+\underset{C}{\overset{D}{\int}}\vec{B}\cdot d\vec{x}+\underset{D}{\overset{A}{\int}}\vec{B}\cdot d\vec{x}=\mu_0 i$$

$$\downarrow\quad\downarrow\quad\downarrow$$

$$0\quad0\quad0$$

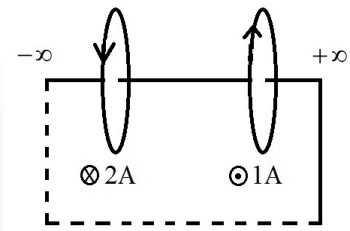
(Large separation)

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$$\int_A^B \vec{B}\cdot d\vec{x} = \mu_0 i$$

For two ring

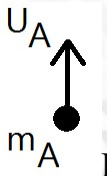


$$\int_{-\infty}^{+\infty} \vec{B}\cdot d\vec{x} = \mu_0(i_{net}) = \mu_0(2-1) \qquad \int_{-\infty}^{+\infty} \frac{\vec{B}\cdot d\vec{x}}{\mu_0} = 1$$

13. $\lambda_A = \lambda_0 + v_s T = vT + \frac{v}{3}T = \frac{4}{3}\frac{v}{f}$

$$\lambda_B = \frac{\frac{4}{3}\frac{v}{f}}{\frac{1}{4}} = \frac{16}{3}\frac{v}{f}$$

14. m_B ● $\qquad V_{com} = \frac{m_A v_A}{(m_A + m_B)} = V_c$

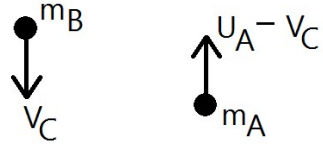


For central impart (perfectly inelastic collision when bond is formed)

$u_{final} = u_{com}$

So, $\frac{m_A u_A = v = 40\,\text{cm/s}}{m_A + m_B} = V = 40\,\text{cm/s}$

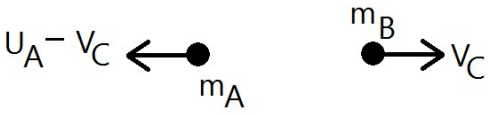
- In com frame →
- Before collision



$P_{net} = m_A(U_A - V_C) - m_B V_C = 0$ [COM is zero momentum frame]

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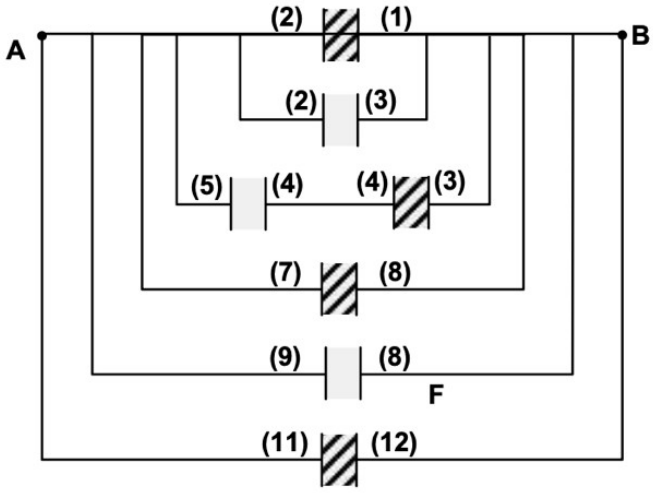
- For off center collision in com frame, we choose a case where B has max radial component.



(For e = 1 in COM frame only direction of velocities may change but their magnitude stays same)

$$t_{min} = \frac{R}{V_C} = \frac{10}{40} = 0.25$$

15,16. Apply Kirchoff's Law on the equivalent circuit shown in the diagram.



17,18. $T = \frac{2\pi}{\sqrt{GM}} r_0^{3/2}$

$$E_1 = \frac{GMm\left(3R^2 - r_L^2\right)}{2R^3} - \frac{GMm}{r_0}$$

$$E_2 = \frac{GMm}{2r_0}$$

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CHEMISTRY

19. a) W (Tungsten) is a d-block element (5d series)
b) W (*n*-pentane) has the highest boiling point among pentane isomers, so is least volatile.
c) Tryptophan is an essential α-amino acid whose one letter code is W
d) W (watt) is the SI unit of power

20. a) acidity order: **cyanoform** > nitroform > chloroform
b) melting point order: **acetamide** > acetic acid > ethyl acetate
c) boiling point order: $GeCl_4 > CCl_4 >$ **$SiCl_4$**
d) bond angle order: $Cl_2O > H_2O >$ **F_2O**

21. a) HN_3 and HCN are weak acids containing sp hybridised atom(s)
b) NaN_3 converts acid chloride to acyl azide in Curtius rearrangement
c) NaCN is a catalyst in Benzoin condensation
d) $\Delta T_f = iK_f m \qquad = 2\times1.86\times\dfrac{12.2}{122} \qquad = 0.372$

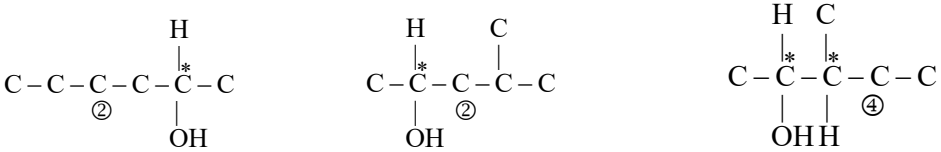
22. a)
$$\frac{3000}{\left(\frac{24}{2}\right)} = \frac{i\times60\times60}{96500}\, i_{\text{amperes}} = \frac{3000\times96500}{12\times60\times60} = \frac{3000\times965}{12\times6\times6} = \frac{250}{6\times6}\times965 = 6.94\times965 < 7\times1000$$
b) 1 part NaCl is mixed with 4 parts $MgCl_2$ for these purposes.
c) due to porcelain hood Cl_2 gas released at anode escapes out without coming in contact with molten magnesium.
d) molten magnesium ($d = 1.74$ g / cm^3) is less dense than molten electrolyte (2.30 g / cm^3) which is why magnesium floats on the surface of the electrolyte.

23. After Ruff degradation followed by oxidation with conc. HNO_3, *D*-Allose, *D*-Altrose and *D*-Gulose form meso-aldaric acids and hence lose optical activity.

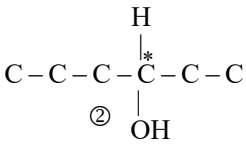
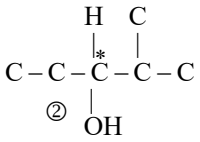
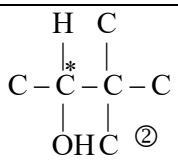
24. a) fact
b) fact
c) fact
d) $Ca + 2N_2O_5 \xrightarrow{\Delta} Ca(NO_3)_2 + 2NO_2(g)$
$5Ca + V_2O_5 \xrightarrow{\Delta} 5CaO + 2V(s)$

25. Nylon–6,6 and Kevlar are polyamides, Dacron and Glyptal are polyesters.

26. a) Refer Unit–6,6.2.3, Class XII NCERT (Metallurgy)
b) Refer Unit-9, 9.9 (B), Class XII NCERT (Amines)
c) Refer osazone test for reducing sugars (Biomolecules)
d) Refer Unit-5,5.4.5, Class XII NCERT (Surface Chemistry)

27. 

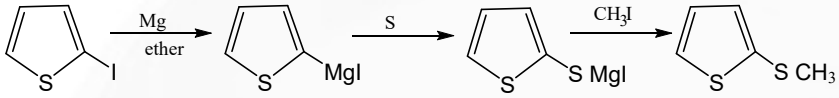
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For the 6 carbon skeleton of 2,3-Dimethylbutane, no secondary alcohol is possible.

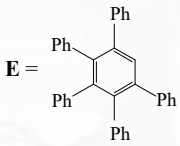
$2+2+2+2+4+2=\mathbf{14}$

28. $K_{a_1}\times K_{a_2}=\dfrac{\left[H^+\right]^2\left[S^{-2}\right]}{\left[H_2S\right]}$ $K_{a_1}\times K_{a_2}=\dfrac{\left[H^+\right]^2}{\left[H_2S\right]}\dfrac{K_{sp}}{\left[Zn^{+2}\right]}10^{-21}=\dfrac{\left[H^+\right]^2}{0.1}\cdot\dfrac{10^{-21}}{10^{-3}}$ $\left[H^+\right]=10^{-2}$

29. 

30. Entropy change accompanying change of phase is given by

$\Delta S_{\text{vap.}}=\dfrac{\Delta H_{\text{vap.}}}{T_{\text{b.p.}}}$ $90=\dfrac{30\times10^3}{T_{bp}}$ $T_{bp}=\dfrac{30\times10^3}{90}=\dfrac{1000}{30}=333.33\ \text{K}$

31. Structure of 

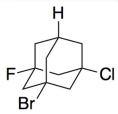
32. $(240\times4)+(32\times8)+(44\times4)+\left(60\times\dfrac{1}{8}\times2\right)-\left[(32\times2)+(44\times1)\right]$

$=(240\times4)+(32\times6)+(44\times3)+\dfrac{120}{8}=1299$

33,34. • P_4O_6 will dissolve and hydrolyse in pure water to form H_3PO_3 which forms HPO_3^{2-} in alkaline aqueous solution

• As_4O_6 will dissolve and hydrolyse in pure water (to a lesser extent than P_4O_6) to form H_3AsO_3 which forms AsO_3^{3-} in alkaline aqueous solution

• $(CH_2)_6N_4$ is soluble in pure water but it does not hydrolyse in pure water or alkaline solution (blood) but readily hydrolyses in acidic solution (urine in urinary bladder) releasing formaldelyde which serves as a urinary antiseptic to treat UTI.

• 

is extremely hydrophobic due to dominance of the hydrocarbon (non-polar) part, hence does not dissolve in pure water. Also due to the rigid cage-like structure adamantane derivatives like these cannot form 2^4 stereoisomers despite containing 4 chiral centres. Only 2 stereoisomers (the one shown above and its mirror image) will exist.

35,36. $T_1=PbS$ $T_2=KI$ $T_3=K_2C_2O_7$ $T_4=HgI_2$

$F_1=O_2$ $F_2=SO_2$ $F_3=H_2S$

$B_1=H_2O_2$ $B_2=H_2SO_4$ $B_3=NH_4OH$

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MATHEMATICS

37. $P\left(\frac{3}{2}, \sqrt{3}\right), Q\left(\frac{3}{2}, -\sqrt{3}\right), R(6, 0), S\left(\frac{-3}{2}, 0\right)$

$L_1\left(\frac{1}{2}, 1\right), L_2\left(\frac{1}{2}, -1\right), L_3\left(\sqrt{5}, \frac{4}{3}\right), L_4\left(\sqrt{5}, \frac{-4}{3}\right)$

38. Conceptual

39. No. of equilateral triangles = 11

No. of isosceles triangles = 120

No. of triangles = 285

40. $P_4 = \frac{{}^4C_2}{4^4}, P_5 = 0$

41. Let roots be $\alpha, \beta, \gamma \in N$

$\alpha + \beta + \gamma = \alpha$

$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{a^2 - 81}{2}$

$\alpha\beta\gamma = \frac{C}{2}$

$\therefore \quad \alpha^2 + \beta^2 + \gamma^2 = 81$

$\therefore \quad (\alpha, \beta, \gamma) = (1, 4, 8), (3,6,6)(4,4,7)$

$a = \alpha + \beta + \gamma = 13, 15, 15$

$C = \alpha\beta\gamma = 64, 216, 226$

42. Conceptual

43. $P\ (1, -1, 0)$

$Q\left(0, \frac{-1}{2}, \frac{3}{2}\right)$

$\alpha = 4$

44. Clearly, $f(-1) = -f(1), f(0) = 0$

For option (A) and (B), apply LMVT for the function $y = f(x)$ in $[-1,0]$ and $[0,1]$

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respectively

For option (D), apply LMVT to the function $y=(f(x))^n$ in $[0,1]$

$f(-1)=-f(1),\ f(1)=0$

Use $LMVT$ on $y=f(x)$ in $[-1,0]$ and $[0,1]$

Use LMVT on $y=(f(x))^n$ in $[0,1]$

45. $\left|\frac{1}{r}\sin x+\frac{1}{r+1}\cos x\right|$ is periodic with period π.

$\therefore$ In $=2\sqrt{\frac{1}{r^2}+\frac{1}{(r+1)^2}}$

46. Given:

$$S_n=\sum_{k\geq 0}\binom{n+k}{2k}2^{n-k}$$

$$2S_n=aS_{n-1}+bS_{n-2}\ (n\geq 5)$$

$$a^2+b^2$$

Step 1: Compute few values:

$S_1=3$

$S_2=11$

$S_3=43$

$S_4=171$

$S_5=683$

$S_6=2731$

Step 2: Recurrence

Use:

$$2S_n=aS_{n-1}+bS_{n-2}$$

For $=5$:

$$2S_5=2\cdot 683=1366$$

$$aS_4+bS_3=171a+43b$$

So:

$171a+43b=1366$

Step 3: Solve equations

Solve (1) and (2):

$171a+43b=1366$

$683a+171b=5462$

Solve

$a=10,\ b=-8$

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Step 4: Final Answer

$a^2+b^2=10^2+(-8)^2=100+64=164$

47. $P(r)=(2r+1)^2$

48. $\dfrac{a_1+a_3+a_5+.....+a_{9n}}{50}=1$

49. α,β are roots of $x^2-5x+2=0$

$\therefore\ \alpha+\beta=9, \qquad \alpha\beta=2$

$$\frac{4\alpha+\beta^5}{10\beta^2}=\frac{\alpha^2\beta^2\alpha+\beta^5}{10\beta^2}$$

$$=\frac{\left(\alpha^3+\beta^3\right)}{10}$$

50. Conceptual

51-52. $f'(x)$ is increasing function $\forall\ x\in(-3,4)$

$$Q'(x)=2xf'\left(\frac{x^2}{3}\right)-2xf'(3-x^2)$$

For $Q'(x)$ to be increasing $Q'(x)>0$

$$2xf'\left(\frac{x^2}{3}\right)>2xf'\left(3-x^2\right)$$

None $\forall\ x\in(-3,0)\Rightarrow f'\left(\dfrac{x^2}{2}\right)<f'\left(3-x^2\right)$

and $\forall\ x\in(0,4)\Rightarrow f'\left(\dfrac{x^2}{2}\right)>f'(3-x^2)$

53-54. $P(x)=(x+1)(x-3a+1)\left(x+\dfrac{a+1}{2}\right)$

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